MTH 331(A) Dr. James Ashe

Exam 4 Review **Name:**

12.1: Vectors in the Plane

- Basic vector operations and arithmetic: **17-20, 39, 41, 43**
  - Magnitude and unit vectors.
  - Adding, subtracting, scalars.
  - Parallelogram/Triangle rule.
- Word problems: **45, 47**
  - Describe various motions as vectors or as a combination of vectors.
  - Find a vector description given the direction and magnitude of a force.
  - Find the magnitude and direction given a vector.

12.2: Vectors in Three Dimensions

- Graphing points in $\mathbb{R}^3$: **9-21**
- Basic vector operations and arithmetic in $\mathbb{R}^3$ (similar to 12.1 above): **35, 37**

12.3: Dot Products

- Compute the dot product, $\mathbf{u} \cdot \mathbf{v}$: **13-17**
- Find the angle between two vectors (in 2 or 3 dimensions): **13-17**
- Find $\text{proj}_{\mathbf{v}}\mathbf{u}$ and $\text{scal}_{\mathbf{v}}\mathbf{u}$: **19-21, 25-27**
- Word problems: **29-31**
  - Find the *work*, given vectors and/or magnitude and directions.

12.4: Cross Products

- Compute the cross product, $\mathbf{u} \times \mathbf{v}$, and find an orthogonal vector: **9-11, 25, 29**
- Find torque: **33, 57**
- Word problems: **37, 38, 57, 58**
  - Describe a resulting force as the cross product of vectors.
  - Find the magnitude and direction of the force.

12.5: Lines and Curves in Space

Continuous motions can be described using vector-valued functions (also called position vector), $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$. This is a parametric function in $\mathbb{R}^3$.

- Concept questions about vector-valued functions: **1-8, 39-41**
- Finding an equation of a line meeting given conditions: **9-13**

12.6: Calculus of Vector-Valued Functions

- Find the derivative of a vector-valued function, $\mathbf{r}(t)$: **7-15, 19, 31**
  - Review the power, product, quotient, and chain rules. Also review the derivative rules for the natural logarithmic, exponential, and trigonometric functions.
- Find the definite and indefinite integral of $\mathbf{r}(t)$: **41-43, 49-53**
  - Review the power and substitution rule. Also review the integration rules for the natural logarithmic, exponential, and trigonometric functions. You will not need to use integration by parts.
- You will be provided all the necessary derivative and integration rules (as they appear in the text).